

EDA
9/20/2026

Missing**Incomplete**

Attempt 1



Review Feedback

Offline Score:

Incomplete

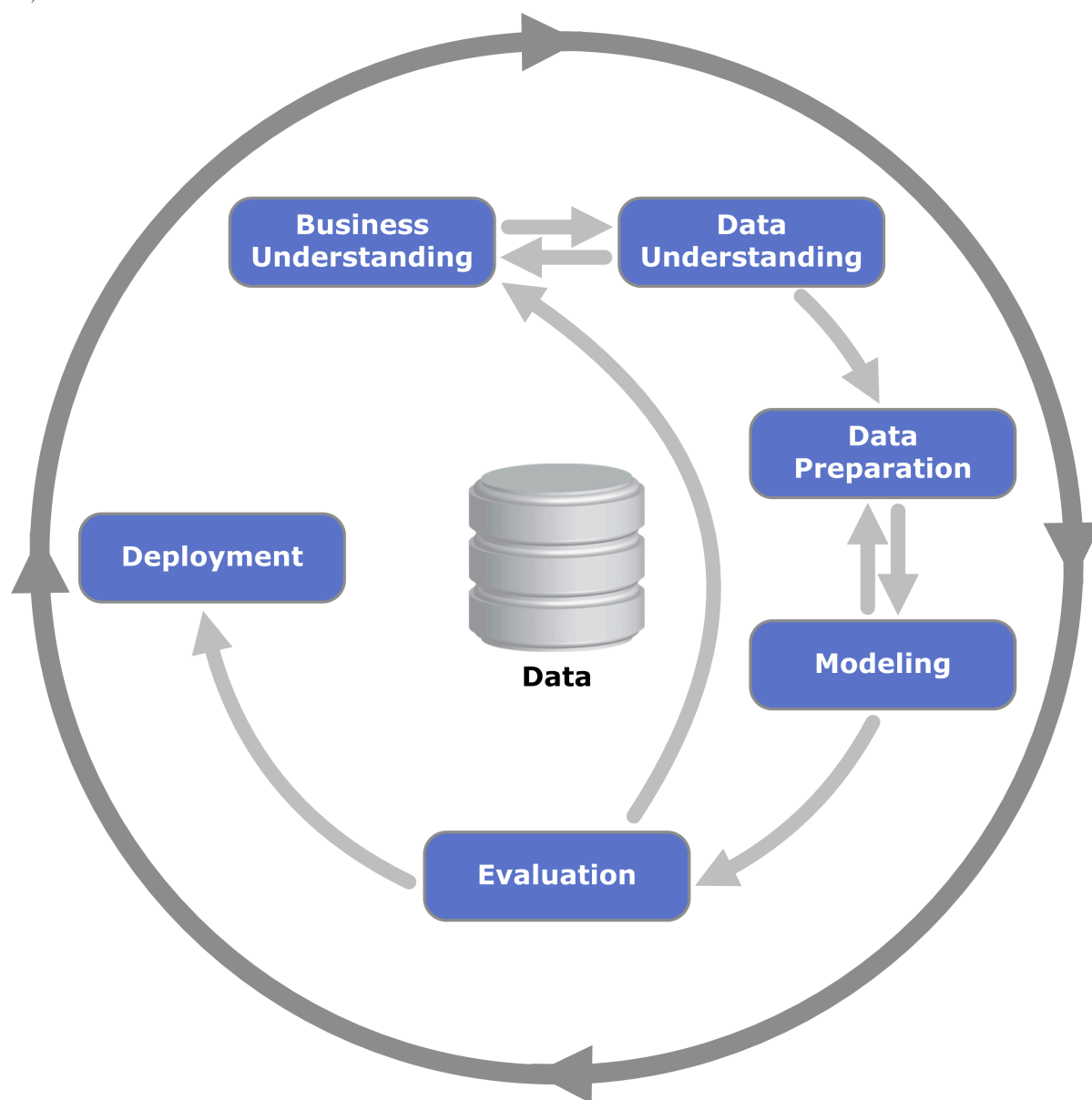
Add Comment

Anonymous Grading: **no****Unlimited Attempts Allowed**

▼ Details

EDA

Exploratory Data Analysis or EDA is a necessary process for any project, involving the data understanding and data preparation CRISP-DM phases.



The assignment is to do EDA for the Home Credit Default Risk project in a notebook that combines code and written interpretation.

Objectives

1. Understand the role of EDA in a project as the bridge from problem framing to data preparation.
2. Practice using data science AI tools to create a detailed Quarto EDA notebook.
3. Learn the structure of the project data and identify its problems (missing values, impossible values, grain and join issues) and decide on solutions.

Assignment

Compile and submit your EDA notebook, which should take the form of an HTML document, with code chunks revealed and warning and lengthy output suppressed.

What are we expecting exactly?

- 1. A well-organized notebook with headings and table of contents.
- 2. A clear statement of the business problem. This information can be borrowed from your business problem statement.
- 3. Data summaries based on your questions about the data: plots and summary tables. Make sure that you include plot titles and axis labels.
- 4. Thoughtful reflection on the meaning and significance of exploratory results--plots and tables--for the project. This interpretive writing belongs in the main sections of the notebook dedicated to data exploration, and should go *between* code chunks.
- 5. Code annotation. This writing goes *within* code chunks, and should explain what the code is doing.
- 6. A results section at the end where you summarize the key points of what you have learned through EDA. Have you discovered any data problems? Do some relationships seem particularly strong? How has your EDA influenced your thinking about your analytics approach?
- 7. Quality, error-free writing.

Note: This stage stage of the project requires only `application_{train|test}.csv`. The supplementary tables (bureau, previous applications, installments, and the rest) are optional.

Please also include a description of how you used AI to produce the notebook. How did it go?

See the scoring rubric for details on grading.

 Resources

- 1. **Exploratory Data Analysis with AI Assistance** (<https://rstudio-connect.business.utah.edu/content/5ee88535-fa59-4074-afd8-acb3a8b9c2e8/eda-with-ai-slides.html>)
- 2. **Applied Predictive Modeling** (<https://utah.instructure.com/courses/1262601/files/202742627?wrap=1>) (APM). See chapter 3 especially.
- 3. **ABOK 8.2.13 - 8.2.19** (<https://utah.instructure.com/courses/1262601/files/202742596?wrap=1>) (Data Understanding and Preparation).
- 4. **Datacamp**. (<https://utah.instructure.com/courses/1262601/pages/datacamp>) Use the courses at Datacamp to fill in gaps in your modeling knowledge and skills.

▼ View Rubric

EDA notebook rubric		
Criteria	Ratings	Points

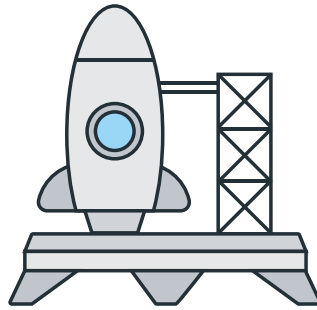
Criteria	Ratings			Points
Table of Contents view longer description	Exemplary TOC present; sections in the required order; headings are the questions. 2 pts	Satisfactory TOC present; some sections missing, misordered, or headed by topic rather than question. 1 pts	Below Expectation No TOC, or structure doesn't follow the assignment. 0 pts	/2 pts
Introduction view longer description	Exemplary Problem stated clearly; questions trace to the BPS, feature report, or data map. 2 pts	Satisfactory Problem stated; questions present but some lack a decision or are description rather than questions. 1 pts	Below Expectation Problem missing or generic; no question list, or questions are the tool's defaults. 0 pts	/2 pts
Description of the data view longer description	Exemplary Application tables fully described; column groups identified; dictionary discrepancies noted; any supplementary table used is described with grain, join, and coverage. 2 pts	Satisfactory Application tables described; column groups or discrepancies missing; supplementary tables used but not mapped. 1 pts	Below Expectation Missing or row counts and column lists only. 0 pts	/2 pts
Discussion of data integrity view longer description	Exemplary All required checks present with test, finding, and decision; missingness examined against the target; at least one finding beyond the well-known ones; temporal check present if joins were made. 2 pts	Satisfactory Most checks present; findings without decisions, or missingness counted but not related to the target. 1 pts	Below Expectation Checks missing or reduced to a missingness chart and the placeholder value everyone knows. 0 pts	/2 pts

Criteria	Ratings			Points
Exploratory results view longer description	Exemplary Every question answered with an appropriate plot or table; titles and axis labels throughout; code annotated. 2 pts	Satisfactory Most questions answered; some plots unlabeled or some output not tied to a question. 1 pts	Below Expectation Plots and tables present but not organized by question, or generic profiling output. 0 pts	/2 pts
Interpretation view longer description	Exemplary Every section interpreted; interpretations name a mechanism and end in a decision; goes beyond restating the output. 2 pts	Satisfactory Most sections interpreted; some restate the output or stop short of a decision. 1 pts	Below Expectation Little interpretation, or interpretation is the tool's one-sentence summaries unchanged. 0 pts	/2 pts
Conclusion view longer description	Exemplary Data problems each paired with a decision; strong predictors named with evidence; explicit statement of what the EDA confirmed or changed in the BPS. 2 pts	Satisfactory Problems and predictors listed; decisions or the BPS reflection thin. 1 pts	Below Expectation Restates the notebook without decisions or reflection. 0 pts	/2 pts
Professional use of notebook view longer description	Exemplary Clean render; no warnings or dataframe dumps; error-free prose. 2 pts	Satisfactory Renders; minor warnings, some long output 1 pts	Below Expectation Doesn't render, long unsuppressed output, or writing with errors throughout. 0 pts	/2 pts

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